

IoT-Based Smart Flood Monitoring and Early Warning System Using ESP32 and Blynk Cloud

Abstract

Floods are among the most common natural disasters, causing serious damage to life and property, especially in low-lying and flood-prone areas. Most existing flood monitoring methods rely on manual checking or visual observation, which is slow and often fails to give people enough time to react safely.

This project presents a Smart Flood Monitoring and Early Warning System built using the ESP32 microcontroller integrated with the Blynk Cloud platform. The system continuously monitors water level, temperature, and humidity using a water level sensor, an HC-SR04 ultrasonic sensor, and a DHT22 sensor. All readings, along with the current flood status, are displayed locally on a 16x2 LCD screen. Based on the water level percentage, the system classifies the situation into NORMAL, WARNING, or DANGER. When the danger threshold is crossed, the system triggers a local alert using an LED and buzzer, and simultaneously sends sensor data to the Blynk cloud dashboard along with an instant mobile notification.

This low-cost and automated approach removes the need for constant manual supervision and provides faster, more reliable flood warnings, making it suitable for deployment in flood-prone residential and rural areas.

Introduction

Floods are one of the most common and damaging natural problems we face, especially during monsoon season. Many times, people don't get any warning before water levels rise, and by the time they realize it, it's already too late to react safely.

This project is a small but useful step toward solving that problem. It is a Smart Flood Monitoring and Alert System that keeps checking water level, temperature, and humidity in real time using sensors connected to an ESP32 board. Whenever the water level becomes dangerous, the system immediately turns on a buzzer and LED for a local warning, and also sends an alert notification to the user's phone through the Blynk app.

A common question that comes up here is why ESP32 was chosen over Arduino, since Arduino is also a widely used and beginner-friendly board. The main reason is connectivity. Arduino boards such as the Arduino Uno do not have built-in Wi-Fi, so an additional Wi-Fi module would be

needed to connect them to the internet, increasing both cost and circuit complexity. ESP32, on the other hand, comes with built-in Wi-Fi, more processing power, and more input/output pins in a single compact board. Since this project depends entirely on sending real-time data to the cloud, ESP32 was a more practical and efficient choice.

In short, this project combines simple, affordable hardware with cloud connectivity to build a flood monitoring system that works both locally and remotely, helping people get timely warnings before a flood situation becomes serious.

problem statement

Floods often strike without warning, causing serious damage to life and property. Most areas still depend on manual checking or visual observation to detect rising water levels, which is slow and unreliable. Traditional flood monitoring systems are usually expensive and require continuous human supervision, making them impractical for small towns and villages. There is a need for an affordable, automated system that can continuously monitor water levels and send instant alerts without relying on manual effort or costly equipment. This project aims to solve this problem using ESP32 and basic IoT components, offering a solution that is significantly more affordable than traditional flood monitoring setups.

Objective

To monitor water level, temperature, and humidity in real time using ESP32 and connected sensors.to classify flood risk into NORMAL, WARNING, and DANGER based on water level percentage.to display live sensor data and flood status on an LCD screen.to trigger local alerts (LED and buzzer) when the danger threshold is reached.to send real-time data to the Blynk cloud dashboard for remote monitoring.to deliver instant mobile alerts during dangerous flood conditions using Blynk notifications.

Components Used

ESP32:-Main controller, handles sensors and Wi-Fi communication

DHT22:-SensorMeasures temperature and humidity

HC-SR04:-Ultrasonic SensorMeasures distance, used to calculate

water level**Water Level Sensor (simulated in Wokwi)**:-Detects water presence/level

16x2 LCD with I2C Module:-Displays live readings on-site

Buzzer:-Gives a sound alert during danger

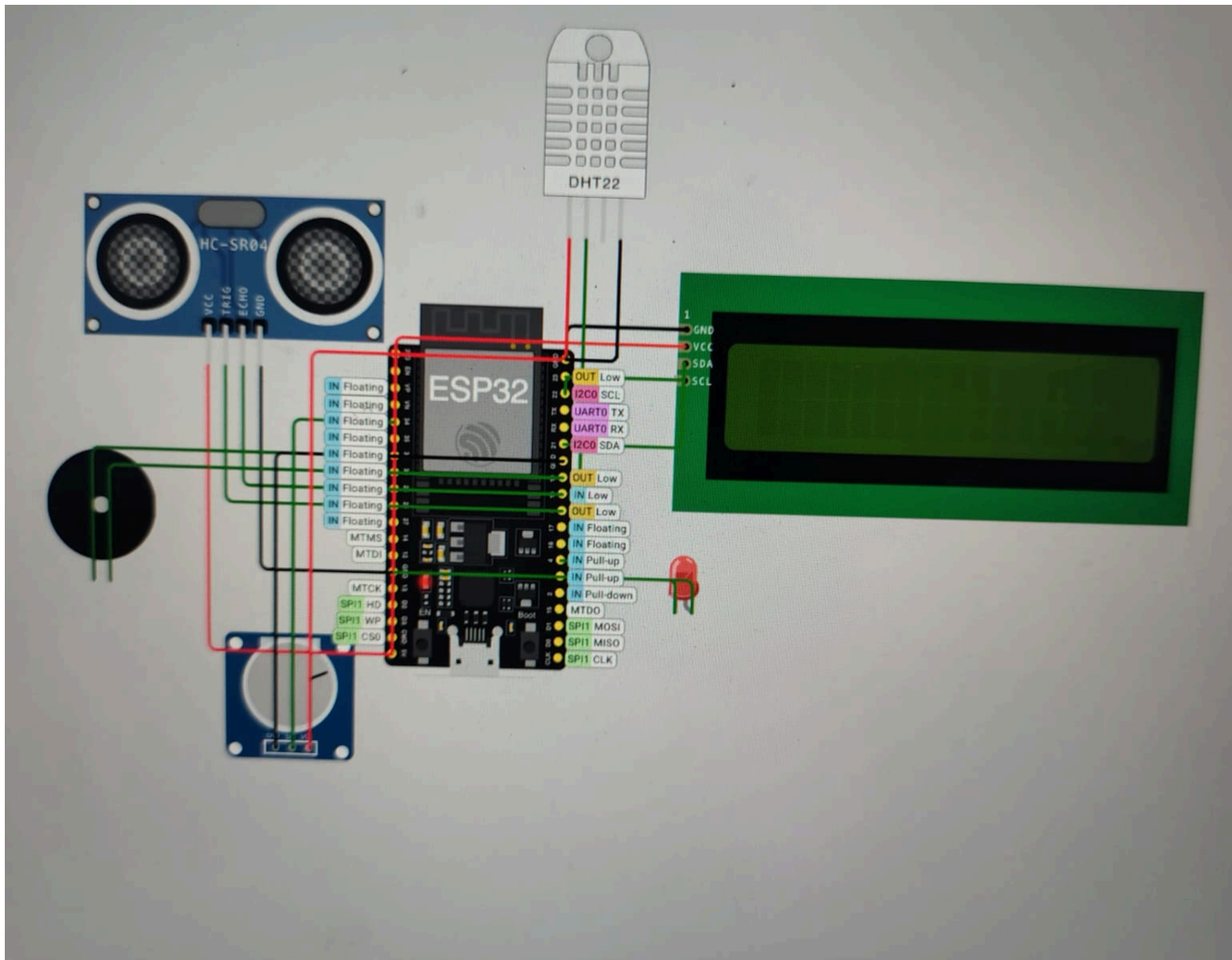
LED:-Gives a visual warning light

Jumper Wires / Breadboard:-Used for connecting all components

Blynk Cloud Platform:-Used for remote dashboard and mobile alerts

Wokwi Simulator:-Used to build and test the circuit virtually

Arduino IDE:-Used to write and upload the ESP32 program



Working

Once powered on, the ESP32 continuously reads data from all connected sensors. The DHT22 measures temperature and humidity, while the HC-SR04 ultrasonic sensor measures distance, which is used to calculate the water level percentage. All these readings, along with the current flood status, are displayed on the LCD screen for anyone monitoring the system on-site.

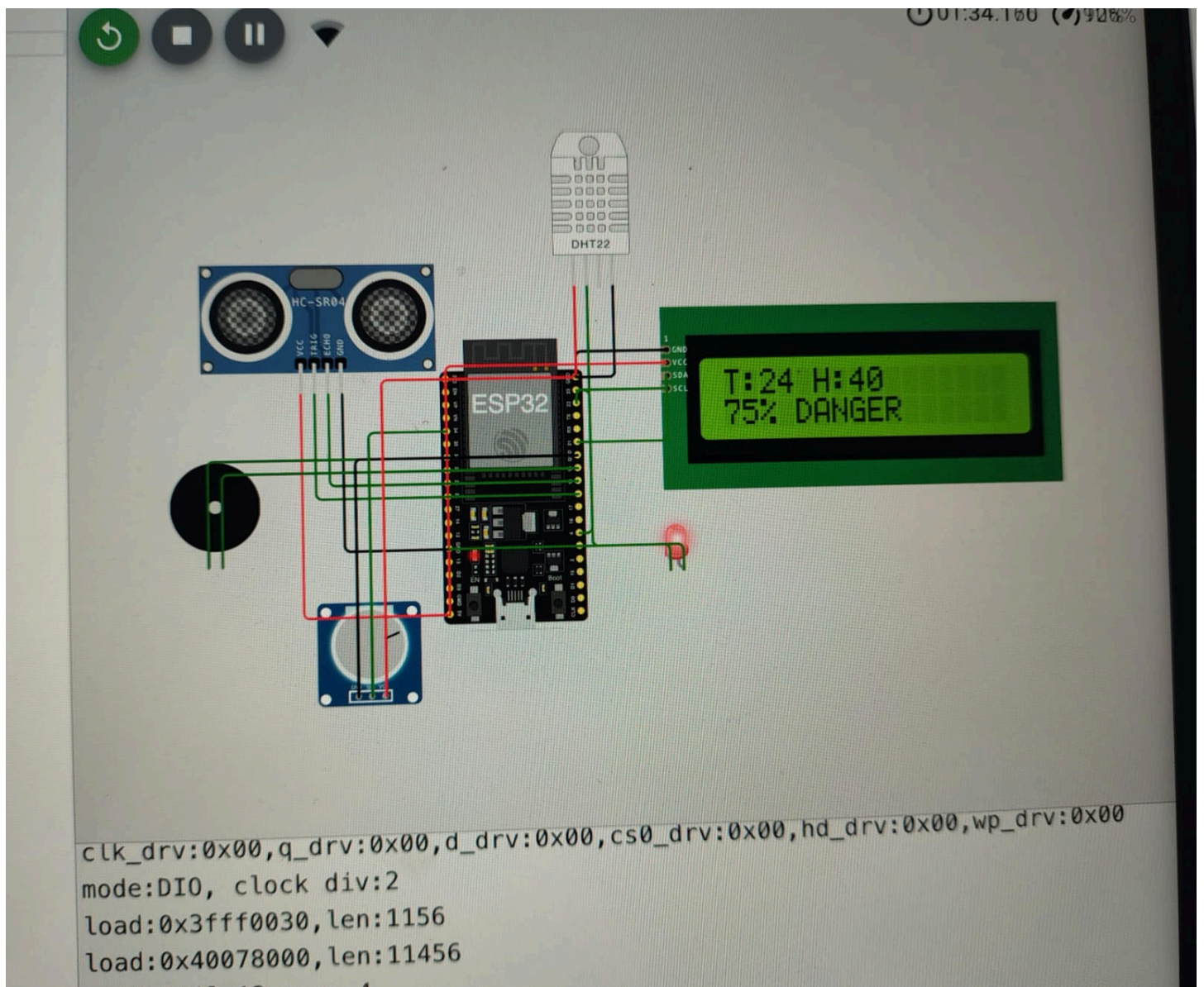
Based on the water level percentage, the system classifies the flood risk into three categories:

Water Level	Status
0% – 40%	NORMAL
41% – 70%	WARNING
Above 70%	DANGER

If the water level enters the DANGER range, the system immediately turns on the LED and buzzer to alert anyone nearby. At the same time, the sensor data is sent to the Blynk cloud dashboard, which displays the following in real time:

- Water Level (%)
- Temperature
- Humidity
- Flood Status
- Distance

When the DANGER level is reached, Blynk also triggers an instant mobile notification using a custom event called "flood_alert", ensuring that the concerned person is warned immediately, even if they are away from the actual location. This combination of local alerts (LED, buzzer) and remote alerts (Blynk notification) makes the system reliable for both on-site and remote monitoring.



Software used

1. **Wokwi Simulator** –circuit simulation, ESP32 setup testing, and virtual sensor integration
Arduino-style ESP32 programming environment –code development and ESP32 programming .
2. **Blynk Cloud Platform** –IoT dashboard creation, cloud data monitoring, and mobile notification integration .
3. **Blynk Mobile App / Web Dashboard** –real-time sensor data visualization and remote flood alert monitoring .

Blynk Dashboard and Mobile Notification Integration

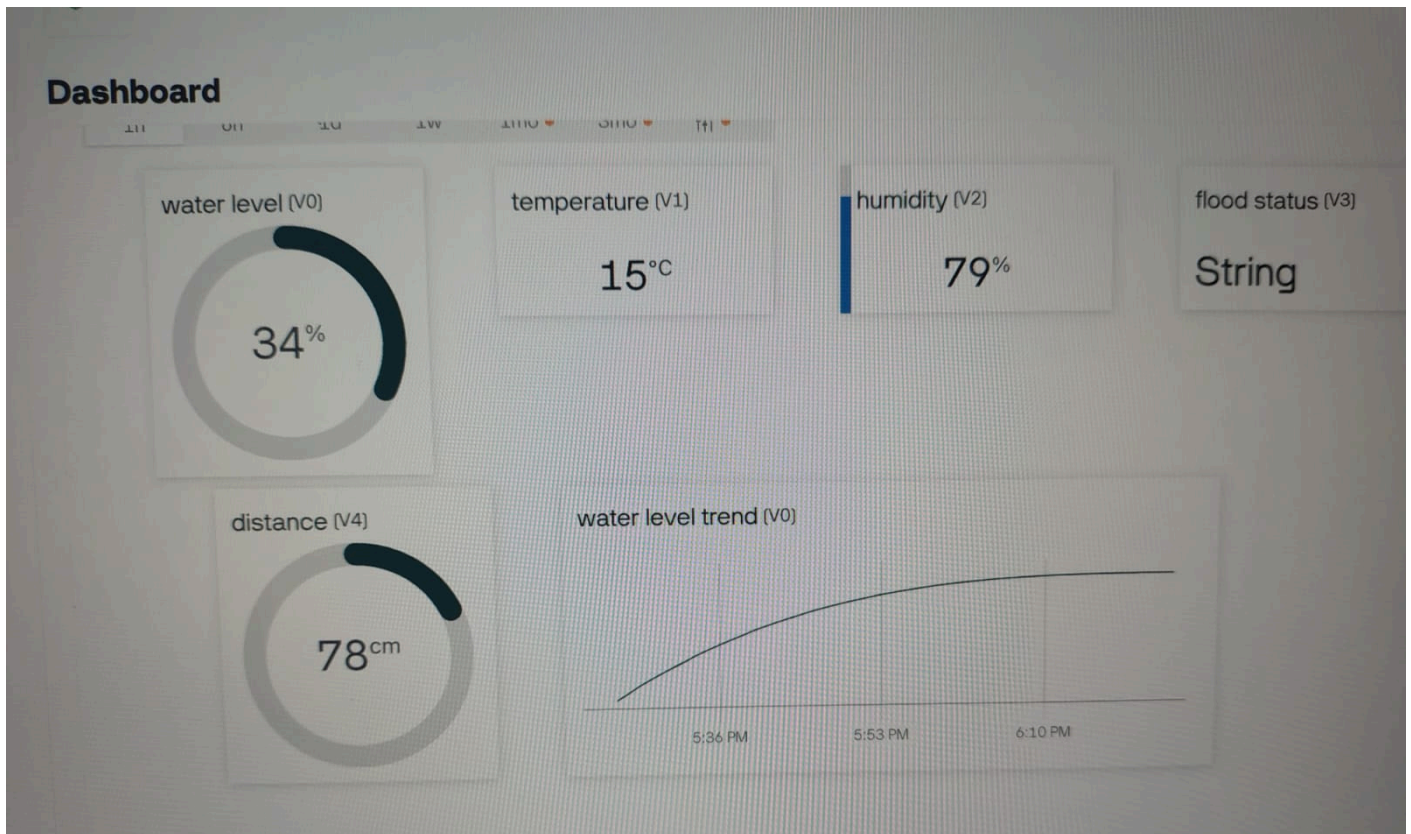
To make the system work remotely, it was connected to the Blynk Cloud platform using ESP32's built-in Wi-Fi. The sensor readings (temperature, humidity, water level, and distance) are sent from the ESP32 to a Blynk dashboard, where they are shown live using simple widgets.

The dashboard displays:

- Water Level %

- Temperature
- Humidity
- Flood Status
- Distance

When the water level crosses the danger threshold, the LED and buzzer turn on as a local alert, and at the same time, Blynk sends a mobile notification (using the event "flood_alert") to warn the user remotely. This way, the project works as both a local and remote flood warning system



Output / Result Analysis

The developed system was successfully simulated and tested using **Wokwi** with **ESP32**, DHT22, ultrasonic sensor, LCD, LED, buzzer, and Blynk cloud integration. The project was able to monitor environmental and water-related parameters in real time and display the output both locally and remotely.

Observed Outputs

1. **Temperature and Humidity Monitoring**

The DHT22 sensor successfully measured temperature and humidity values, which were displayed on the LCD and also transmitted to the Blynk dashboard.

2. **Water Level Percentage Monitoring**

The system continuously read the water input and converted it into a percentage value. This water level percentage was displayed on the LCD and sent to the Blynk dashboard for remote monitoring.

3. Distance Measurement Using Ultrasonic Sensor

The HC-SR04 ultrasonic sensor measured distance values, which helped simulate water-level-related environmental conditions in the project. The measured distance was also displayed on the dashboard.

4. Flood Status Detection

Based on the water level percentage, the system correctly classified the flood condition into three categories:

- **NORMAL** for safe water levels
- **WARNING** for moderate rise in water level
- **DANGER** when the water level crossed the critical threshold

5. Local Alert Mechanism

When the water level exceeded the danger threshold, the **LED** and **buzzer** were activated successfully, indicating a local warning condition.

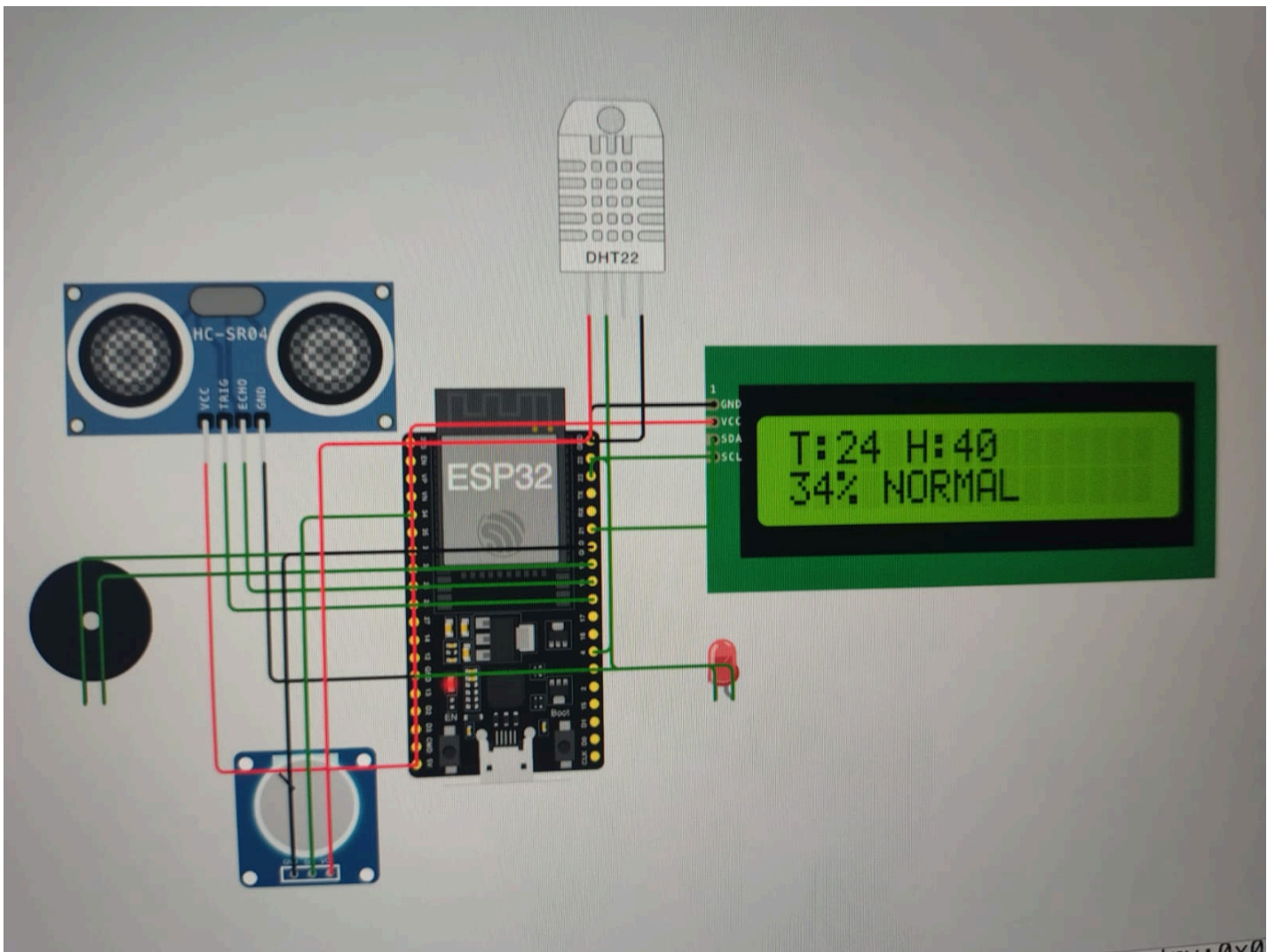
6. Remote Dashboard Monitoring

The Blynk dashboard successfully displayed the following live parameters:

- Water Level %
- Temperature
- Humidity
- Flood Status
- Distance

7. Mobile Notification Alert

During danger conditions, the system triggered the flood_alert event and sent a mobile notification through Blynk, demonstrating the remote alert capability of the project.



Advantages

- 1. Real-time flood monitoring**
The system continuously monitors water level, temperature, humidity, and distance in real time.
- 2. Remote monitoring through Blynk dashboard**
Sensor data can be viewed from anywhere using the Blynk cloud dashboard, making the system more accessible and user-friendly.
- 3. Early warning and alert system**
When the water level crosses the danger threshold, the system activates buzzer and LED alerts and can also send a mobile notification through Blynk.
- 4. IoT-based smart integration**
By integrating ESP32 with Blynk, the project combines embedded systems with IoT technology, making it more advanced than a basic standalone flood alert system.
- 5. Scalable for future smart city or disaster monitoring systems**
The system can be extended with more sensors, cloud storage, analytics, and advanced alert mechanisms.

Limitations

1. **Prototype-level implementation**

The project is currently developed as a simulation/prototype, so it may require additional testing for real-world deployment.

2. **Sensor accuracy limitations**

Readings from water level, ultrasonic, and DHT22 sensors may vary due to environmental noise, placement, or calibration issues.

3. **Dependence on internet connectivity**

Blynk dashboard monitoring and mobile notifications require a stable Wi-Fi/internet connection.

4. **Power dependency**

The system needs continuous power supply to function properly, which may be a challenge during extreme flood situations.

5. **Basic threshold-based decision logic**

Flood status is currently determined using fixed thresholds (Normal, Warning, Danger), which may not fully represent complex real-world flood conditions.

Future scope

- Add GSM/SMS alerts for areas without Wi-Fi
- Integrate cloud data logging and analytics
- Use AI/ML for flood prediction and trend analysis
- Deploy solar-powered outdoor version for real-world use
- Add mobile app with live monitoring and emergency response features

Conclusion

This project successfully demonstrates how a simple, low-cost IoT setup using ESP32 and Blynk Cloud can be used to build an effective flood monitoring and early warning system. By combining real-time sensor monitoring with local alerts and cloud-based remote notifications, the system addresses the key problem of delayed flood warnings, especially in areas that lack access to expensive monitoring infrastructure. The project shows strong potential for further enhancement, such as adding GSM-based SMS alerts, solar power for remote deployment, and predictive flood modeling, making it a solid foundation for future smart disaster management solutions.